

[D.3.2] Universal AI: AIXI

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Iliad Intensive

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- **The common engine**: a single **Bayesian mixture** ξ over a class $\mathcal{M}$ of candidate environments, weighted by a prior w_ν .
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- **Universal** choice: $\mathcal{M}$ = all computable environments, prior $w_\nu = 2^{-K(\nu)}$ (K is the “complexity” of the environment). Assumption “ $\mu \in \mathcal{M}$ ” becomes “*the universe is computable*”.

The Bayesian mixture ξ

Definition

Class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ with prior weights $w_\nu > 0$, $\sum_\nu w_\nu = 1$.

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- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \nu(x_t \mid x_{<t}), \quad w(\nu \mid x_{<t}) = w_\nu \frac{\nu(x_{<t})}{\xi(x_{<t})}.$$

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- **Posterior update is multiplicative:** $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \frac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$: we reweight by how well ν predicted vs. ξ (Bayes' rule!)

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- Spaces: actions $\mathcal{A}$, observations $\mathcal{O}$, rewards $\mathcal{R} \subset [0, 1]$, percepts $\mathcal{E} = \mathcal{O} \times \mathcal{R}$, and (finite) histories $\mathcal{H}^* := (\mathcal{A} \times \mathcal{E})^* := \cup_{t=0}^{\infty} (\mathcal{A} \times \mathcal{E})^t$.

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- **No Markov / MDP assumption!** Both the policy and environment can depend on all prior percepts.

The three players

Policy π

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Interaction measure ν^π

Run π inside ν and multiply along the history:

$$\nu^\pi(\mathfrak{ae}_{t:m} \mid \mathfrak{ae}_{<t}) := \prod_{k=t}^m \pi(a_k \mid \mathfrak{ae}_{<k}) \nu(e_k \mid \mathfrak{ae}_{<k} a_k).$$

Value, optimality, and AIXI

Value function (assuming geometric discount γ)

$$\begin{aligned} V_{\nu}^{\pi,m}(\mathfrak{a}_{<t}) &= (1 - \gamma) \mathbb{E}_{\nu}^{\pi}[G_{t-1:m} \mid \mathfrak{a}_{<t}] \text{ with } G_{t-1:m} = \sum_{k=t}^m \gamma^{k-t} r_k \\ &= (1 - \gamma) \sum_{\mathfrak{a}_{t:m}} \nu^{\pi}(\mathfrak{a}_{t:m} \mid \mathfrak{a}_{<t}) \sum_{k=t}^m \gamma^{k-t} r_k \end{aligned}$$

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- **Optimal value** $V_{\nu}^* := \sup_{\pi} V_{\nu}^{\pi}$; an optimal policy π_{ν}^* satisfies $V_{\nu}^{\pi_{\nu}^*} = V_{\nu}^*$.
- **Bayes-optimal policy** $\pi_{\xi}^* \in \arg \max_{\pi} V_{\xi}^{\pi}$: acts optimally w.r.t. the *mixture* ξ .

AIXI

π_{ξ}^* with $\mathcal{M} = \text{all } \approx\text{computable environments}^a$ and prior $w_{\nu} = 2^{-K(\nu)}$. Write ξ_U for ξ with this choice of $\mathcal{M}$ and w_{ν} , so $\text{AIXI} = \pi_{\xi_U}^*$.

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AIXI, explicitly

max over actions, $\sum$ over percepts, choosing actions that maximise the ξ -weighted discounted return:

$$a_t = \underset{a_t}{\text{arg max}} \lim_{m \rightarrow \infty} \sum_{e_t} \max_{a_{t+1}} \sum_{e_{t+1}} \cdots \max_{a_m} \sum_{e_m} \xi(e_{t:m} \mid \mathfrak{x}_{<t} a_{t:m}) \sum_{k=t}^m \gamma^{k-t} r_k$$

Result 1: On-policy value convergence

Theorem

For any $\mu \in \mathcal{M}$ and any policy π ,

$$V_{\xi}^{\pi}(\mathfrak{a}_{<t}) - V_{\mu}^{\pi}(\mathfrak{a}_{<t}) \rightarrow 0 \quad \mu^{\pi}\text{-almost surely.}$$

Intuition: *Following policy π , pretending the environment is ξ , eventually will act as if we knew the true environment μ .*

Result 2: AIXI can't be fooled

Theorem (deterministic μ)

If $V_{\mu}^*(\mathfrak{a}_{<t}) > \varepsilon$ along history $\mathfrak{a}_{<t}$, then

$$V_{\xi}^*(\mathfrak{a}_{<t}) \geq w_{\mu} \varepsilon > 0.$$

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Intuition: *Whenever good behaviour in μ is achievable, the Bayes-optimal agent secures at least a fraction proportional to the prior weight of μ .*

Result 3: Self-optimizing property (stretch goal)

Theorem

If *some* policy can learn to act optimally in every $\nu \in \mathcal{M}$, then π_ξ^* does too:

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Intuition: *AIXI learns to play as well as the optimal agent that knows the true environment μ .*

Takeaways

- **Solomonoff** solves induction: mix over all computable hypotheses, weighted by simplicity. Total prediction error $\leq K(\mu) \ln 2$.

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- **Why care for safety:** a precise, assumption-light model of idealized intelligence providing a lens on what optimal agents do. For example, we can show under what circumstances AIXI would [wirehead](#) [RO11],
- **Approximations:** [MC-AIXI-CTW](#) [VNH⁺11] gives an approximation using Context Tree Weighting to replace ξ and Monte-Carlo-Tree-Search to replace the expectimax. [safely rather than recklessly explore](#) [CCH20], or [self-modify](#) [OR11] as a model of what superintelligent agents would do.

References I

- [CCH20] Michael K. Cohen, Elliot Catt, and Marcus Hutter.
Curiosity killed or incapacitated the cat and the asymptotically optimal agent.
CoRR, abs/2006.03357, 2020.
URL: <https://arxiv.org/abs/2006.03357>.
- [OR11] Laurent Orseau and Mark Ring.
Self-modification and mortality in artificial agents.
In *Artificial General Intelligence (AGI 2011)*, volume 6830 of *Lecture Notes in Computer Science*, pages 1–10.
Springer, 2011.
URL: https://link.springer.com/chapter/10.1007/978-3-642-22887-2_1, doi:10.1007/978-3-642-22887-2_1.
- [RO11] Mark Ring and Laurent Orseau.
Delusion, survival, and intelligent agents.
In *Artificial General Intelligence (AGI 2011)*, volume 6830 of *Lecture Notes in Computer Science*, pages 11–20.
Springer, 2011.
URL: <https://people.idsia.ch/~ring/AGI-2011/Paper-B.pdf>, doi:10.1007/978-3-642-22887-2_2.

References II

- [VNH⁺11] Joel Veness, Kee Siong Ng, Marcus Hutter, William Uther, and David Silver.
A Monte-Carlo AIXI approximation.
Journal of Artificial Intelligence Research, 40:95–142, 2011.
URL: <https://arxiv.org/abs/0909.0801>, doi:10.1613/jair.3125.